

ISE-201: Evaluation of Financial Value of a Bachelor's Degree

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Introduction

Earning a bachelor's degree is a significant commitment, requiring 3-4 years of focused learning in a specific field. For many, securing a job afterward is the ultimate goal. However, before considering to enroll in a bachelors program, certain questions are bound to pass one's mind.

1. Do these jobs offer a return on the investment made in education?
2. Can a degree guarantee financial success?
3. Which fields tend to be more economically rewarding?
4. Moreover, are these high-paying jobs truly dependent on having a college degree?
5. Additionally, what is the gender distribution in the bachelors program and top-earning positions?
6. How do high-paying jobs compare to lower-paying roles within the same industry?

It is essential to analyze data and gain insights to make an informed decision regarding college education, considering the significant time and money involved. With this project, I aim to explore the impact of college education on financial success.

Data

Data Source

This dataset is from FiveThirtyEight datasets in Github.

FiveThirtyEight. (2014). College Majors. GitHub.
CollegeMajorsData

Data collection

The data comes from American Community Survey 2010-2012 Public Use Microdata Series. It was gathered to gain insight on the importance of college education and its influence on the financial outcomes, focusing on factors like the employment rate and earnings after graduating from each major. The cases present in this dataset are purely from an observational study as it is solely based on the survey conducted for graduates from different majors. There were no assumptions or manipulation of data from the researchers while compiling this dataset.

Unit of observation

This dataset has a total of 173 observations where each observation captures information on the number of graduates, employment rate and pay scale for graduates coming out of different majors.

Variables

The dataset consists of 21 variables. Below is a brief description of each of the variable.

```
data <- data.frame(
  Variable = c("Rank", "Major_code", "Major", "Total", "Men", "Women", "Major_category", "ShareW
  Description = c("Rank based on median earnings",
    "Major Code",
    "Major Name",
    "Total number of graduates in the major",
    "Male graduates",
    "Female graduates",
    "Category of Major",
    "Women as share of total",
    "Sample size for the survey conducted",
    "Number of employed graduates",
    "Number of graduates with full time role (>35 hours)",
    "Number of graduates with part time role (<35 hours)",
    "Number of graduates that were employed for atleast 50 weeks in a year and atleast 35
    "Number of graduates that were unemployed",
    "Rate of unemployment for the major",
    "Median earnings after graduation",
    "25th percentile earnings after graduation",
    "75th percentile earnings after graduation",
    "Number of jobs under the major that require college degree",
    "Number of jobs under this major that do not require a college degree",
    "Number of graduates in low wage service jobs"
  )
)

# Display table
knitr::kable(data, caption = "Variables and their Description in College Majors Dataset")
```

Table 1: Variables and their Description in College Majors Dataset

Variable	Description
Rank	Rank based on median earnings
Major_code	Major Code
Major	Major Name
Total	Total number of graduates in the major
Men	Male graduates
Women	Female graduates
Major_category	Category of Major
ShareWomen	Women as share of total
Sample_size	Sample size for the survey conducted
Employed	Number of employed graduates
Full_time	Number of graduates with full time role (>35 hours)
Part_time	Number of graduates with part time role (<35 hours)
Full_time_year_role	Number of graduates that were employed for atleast 50 weeks in a year and atleast 35 working hours in a week
Unemployed	Number of graduates that were unemployed

Variable	Description
Unemployment_rate	Rate of unemployment for the major
Median	Median earnings after graduation
P25th	25th percentile earnings after graduation
P75th	75th percentile earnings after graduation
College_jobs	Number of jobs under the major that require college degree
Non_college_jobs	Number of jobs under this major that do not require a college degree
Low_wage_jobs	Number of graduates in low wage service jobs

A subset of the variables available in this dataset would be used for analysis. Below variables are of key relevance to the questions posed in this study:

1. Major_category
2. ShareWomen
3. Unemployment_rate
4. Median
5. P25th
6. P75th
7. College_jobs
8. Non_college_jobs

Major_code is a variable that has no significant relevancy to the analysis conducted and hence will be removed from the study under the Data Cleanup section.

Data Cleanup

This section checks for data quality issues, missing data, duplicates, format issues etc. and the necessary data quality improvements are performed.

Including Libraries

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    3.5.1      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

## Warning: package 'corrplot' was built under R version 4.4.2

## corrplot 0.95 loaded
```

```
indata <- read.csv("recent-grads.csv")
```

```
dim(indata)
```

Import Dataset

```
## [1] 173 21
```

```
head(indata)
```

```
## Rank Major_code Major Total Men Women
## 1 1 2419 PETROLEUM ENGINEERING 2339 2057 282
## 2 2 2416 MINING AND MINERAL ENGINEERING 756 679 77
## 3 3 2415 METALLURGICAL ENGINEERING 856 725 131
## 4 4 2417 NAVAL ARCHITECTURE AND MARINE ENGINEERING 1258 1123 135
## 5 5 2405 CHEMICAL ENGINEERING 32260 21239 11021
## 6 6 2418 NUCLEAR ENGINEERING 2573 2200 373
## Major_category ShareWomen Sample_size Employed Full_time Part_time
## 1 Engineering 0.1205643 36 1976 1849 270
## 2 Engineering 0.1018519 7 640 556 170
## 3 Engineering 0.1530374 3 648 558 133
## 4 Engineering 0.1073132 16 758 1069 150
## 5 Engineering 0.3416305 289 25694 23170 5180
## 6 Engineering 0.1449670 17 1857 2038 264
## Full_time_year_round Unemployed Unemployment_rate Median P25th P75th
## 1 1207 37 0.01838053 110000 95000 125000
## 2 388 85 0.11724138 75000 55000 90000
## 3 340 16 0.02409639 73000 50000 105000
## 4 692 40 0.05012531 70000 43000 80000
## 5 16697 1672 0.06109771 65000 50000 75000
## 6 1449 400 0.17722641 65000 50000 102000
## College_jobs Non_college_jobs Low_wage_jobs
## 1 1534 364 193
## 2 350 257 50
## 3 456 176 0
## 4 529 102 0
## 5 18314 4440 972
## 6 1142 657 244
```

The data from the csv file is read successfully and all the 173 observations across 21 features are captured.

```
is.data.frame(indata)
```

Dataset Formatting

```
## [1] TRUE
```

Since the data is currently stored as data frame, let's convert it to a tibble for ease of use with tidyverse library.

```
indata <- as_tibble(indata)
is_tibble(indata)
```

```
## [1] TRUE
```

```
summary(indata)
```

Missing Values Check

```
##      Rank      Major_code      Major      Total
## Min.   : 1    Min.   :1100    Length:173    Min.   : 124
## 1st Qu.: 44    1st Qu.:2403    Class :character 1st Qu.: 4550
## Median : 87    Median :3608    Mode  :character Median : 15104
## Mean   : 87    Mean   :3880                    Mean   : 39370
## 3rd Qu.:130    3rd Qu.:5503                    3rd Qu.: 38910
## Max.   :173    Max.   :6403                    Max.   :393735
##                                     NA's   :1
##      Men      Women      Major_category      ShareWomen
## Min.   : 119    Min.   : 0    Length:173    Min.   :0.0000
## 1st Qu.: 2178    1st Qu.: 1778    Class :character 1st Qu.:0.3360
## Median : 5434    Median : 8386    Mode  :character Median :0.5340
## Mean   : 16723    Mean   : 22647                    Mean   :0.5222
## 3rd Qu.: 14631    3rd Qu.: 22554                    3rd Qu.:0.7033
## Max.   :173809    Max.   :307087                    Max.   :0.9690
## NA's   :1        NA's   :1        NA's   :1
## Sample_size      Employed      Full_time      Part_time
## Min.   : 2.0      Min.   : 0      Min.   : 111    Min.   : 0
## 1st Qu.: 39.0      1st Qu.: 3608    1st Qu.: 3154    1st Qu.: 1030
## Median : 130.0      Median : 11797    Median : 10048    Median : 3299
## Mean   : 356.1      Mean   : 31193    Mean   : 26029    Mean   : 8832
## 3rd Qu.: 338.0      3rd Qu.: 31433    3rd Qu.: 25147    3rd Qu.: 9948
## Max.   :4212.0      Max.   :307933    Max.   :251540    Max.   :115172
##
## Full_time_year_round      Unemployed      Unemployment_rate      Median
## Min.   : 111      Min.   : 0      Min.   :0.00000    Min.   : 22000
## 1st Qu.: 2453      1st Qu.: 304      1st Qu.:0.05031    1st Qu.: 33000
## Median : 7413      Median : 893      Median :0.06796    Median : 36000
## Mean   : 19694      Mean   : 2416      Mean   :0.06819    Mean   : 40151
## 3rd Qu.: 16891      3rd Qu.: 2393      3rd Qu.:0.08756    3rd Qu.: 45000
## Max.   :199897      Max.   :28169      Max.   :0.17723    Max.   :110000
##
##      P25th      P75th      College_jobs      Non_college_jobs
## Min.   :18500    Min.   : 22000    Min.   : 0      Min.   : 0
## 1st Qu.:24000    1st Qu.: 42000    1st Qu.: 1675    1st Qu.: 1591
## Median :27000    Median : 47000    Median : 4390    Median : 4595
## Mean   :29501    Mean   : 51494    Mean   : 12323    Mean   : 13284
## 3rd Qu.:33000    3rd Qu.: 60000    3rd Qu.: 14444    3rd Qu.: 11783
## Max.   :95000    Max.   :125000    Max.   :151643    Max.   :148395
##
## Low_wage_jobs
```

```
## Min.    :    0
## 1st Qu.:  340
## Median : 1231
## Mean   : 3859
## 3rd Qu.: 3466
## Max.   :48207
##
```

From the above summary, it is observed that the dataset has one missing value in Total, Men and Women features. These missing values can be filled up to avoid loss of data. The missing values can be filled in with the average values of the major category that it belongs to.

```
indata_cleaned <- indata %>%group_by(Major_category) %>%
  mutate(
    mean_total = round(mean(Total, na.rm = TRUE)),
    mean_men = round(mean(Men, na.rm = TRUE)),
    mean_women = round(mean(Women, na.rm = TRUE)),
    mean_womenshare = mean(Women, na.rm = TRUE)/mean(Total, na.rm = TRUE)
  ) %>%
  mutate(
    Total = case_when(
      is.na(Total) ~ mean_total,
      TRUE ~ Total),
    Men = case_when(
      is.na(Men) ~ mean_men,
      TRUE ~ Men ),
    Women = case_when(
      is.na(Women) ~ mean_women,
      TRUE ~ Women),
    ShareWomen = case_when(
      is.na(ShareWomen) ~ mean_womenshare,
      TRUE ~ ShareWomen),
  ) %>%
  select(-mean_total, -mean_men, -mean_women) %>% ungroup()

summary(indata_cleaned)
```

```
##      Rank      Major_code      Major      Total
## Min.   :  1   Min.   :1100   Length:173   Min.   :  124
## 1st Qu.: 44   1st Qu.:2403   Class :character   1st Qu.:  4626
## Median : 87   Median :3608   Mode  :character   Median : 15058
## Mean   : 87   Mean   :3880                   Mean   : 39191
## 3rd Qu.:130   3rd Qu.:5503                   3rd Qu.: 38844
## Max.   :173   Max.   :6403                   Max.   :393735
##      Men      Women      Major_category      ShareWomen
## Min.   : 119   Min.   :    0   Length:173   Min.   :0.0000
## 1st Qu.: 2200   1st Qu.: 1784   Class :character   1st Qu.:0.3397
## Median : 5347   Median : 8284   Mode  :character   Median :0.5323
## Mean   : 16653   Mean   : 22538                   Mean   :0.5219
## 3rd Qu.: 14440   3rd Qu.: 22456                   3rd Qu.:0.7020
## Max.   :173809   Max.   :307087                   Max.   :0.9690
## Sample_size      Employed      Full_time      Part_time
## Min.   :  2.0   Min.   :    0   Min.   :  111   Min.   :    0
## 1st Qu.: 39.0   1st Qu.: 3608   1st Qu.: 3154   1st Qu.: 1030
```

```
## Median : 130.0    Median : 11797    Median : 10048    Median : 3299
## Mean   : 356.1    Mean   : 31193    Mean   : 26029    Mean   : 8832
## 3rd Qu.: 338.0    3rd Qu.: 31433    3rd Qu.: 25147    3rd Qu.: 9948
## Max.   :4212.0    Max.   :307933    Max.   :251540    Max.   :115172
## Full_time_year_round    Unemployed    Unemployment_rate    Median
## Min.   : 111          Min.   : 0      Min.   :0.00000    Min.   : 22000
## 1st Qu.: 2453          1st Qu.: 304    1st Qu.:0.05031    1st Qu.: 33000
## Median : 7413          Median : 893    Median :0.06796    Median : 36000
## Mean   : 19694          Mean   : 2416    Mean   :0.06819    Mean   : 40151
## 3rd Qu.: 16891          3rd Qu.: 2393    3rd Qu.:0.08756    3rd Qu.: 45000
## Max.   :199897          Max.   :28169    Max.   :0.17723    Max.   :110000
##      P25th      P75th      College_jobs    Non_college_jobs
## Min.   :18500    Min.   : 22000    Min.   : 0      Min.   : 0
## 1st Qu.:24000    1st Qu.: 42000    1st Qu.: 1675    1st Qu.: 1591
## Median :27000    Median : 47000    Median : 4390    Median : 4595
## Mean   :29501    Mean   : 51494    Mean   : 12323    Mean   : 13284
## 3rd Qu.:33000    3rd Qu.: 60000    3rd Qu.: 14444    3rd Qu.: 11783
## Max.   :95000    Max.   :125000    Max.   :151643    Max.   :148395
## Low_wage_jobs    mean_womenshare
## Min.   : 0      Min.   :0.2405
## 1st Qu.: 340    1st Qu.:0.4663
## Median : 1231    Median :0.5154
## Mean   : 3859    Mean   :0.5393
## 3rd Qu.: 3466    3rd Qu.:0.6237
## Max.   :48207    Max.   :0.8370
```

As observed above, the missing values are filled successfully.

Remove Extra Columns Since, Major_code is irrelevant to our data analysis, the column can be removed from the dataset.

```
indata_cleaned <- select(indata_cleaned, -(Major_code))
head(indata_cleaned)
```

```
## # A tibble: 6 x 21
##   Rank Major    Total    Men Women Major_category ShareWomen Sample_size Employed
##   <int> <chr>    <dbl> <dbl> <dbl> <chr>          <dbl>      <int>    <int>
## 1     1 PETROL~   2339  2057   282 Engineering    0.121        36     1976
## 2     2 MINING~    756   679    77 Engineering    0.102         7      640
## 3     3 METALL~    856   725   131 Engineering    0.153         3      648
## 4     4 NAVAL ~   1258  1123   135 Engineering    0.107        16      758
## 5     5 CHEMIC~  32260 21239 11021 Engineering    0.342       289    25694
## 6     6 NUCLEA~   2573  2200   373 Engineering    0.145        17     1857
## # i 12 more variables: Full_time <int>, Part_time <int>,
## #   Full_time_year_round <int>, Unemployed <int>, Unemployment_rate <dbl>,
## #   Median <int>, P25th <int>, P75th <int>, College_jobs <int>,
## #   Non_college_jobs <int>, Low_wage_jobs <int>, mean_womenshare <dbl>
```

```
duplicates_summary <- indata_cleaned %>%
  group_by(Major) %>%
```

```

summarize(count = n()) %>%
filter(count > 1)

print(duplicates_summary)

```

Remove Duplicate Occurences

```

## # A tibble: 0 x 2
## # i 2 variables: Major <chr>, count <int>

```

Since there are no duplicates of Majors in the dataset, no updates are required to be made to the dataset.

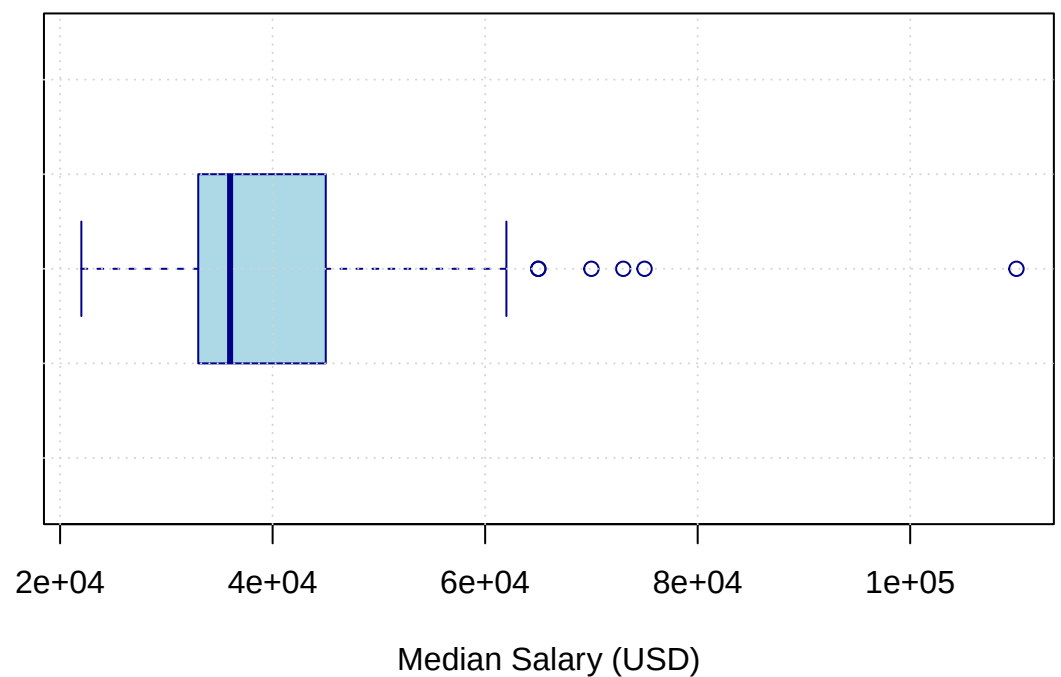
Assess Outliers

```

boxplot(indata_cleaned$Median,
  main = "Box Plot of Median Salary by Major",
  xlab = "Median Salary (USD)",
  col = "lightblue",
  border = "darkblue",
  horizontal = TRUE)
grid()

```

Box Plot of Median Salary by Major



Median Salary


```

median_outliers <- boxplot.stats(indata_cleaned$Median)$out
median_outlier_majors <- indata_cleaned[indata_cleaned$Median %in% median_outliers, c("Major", "Median")]
median_outlier_majors

```

```

## # A tibble: 6 x 2
##   Major                Median
##   <chr>              <int>
## 1 PETROLEUM ENGINEERING 110000
## 2 MINING AND MINERAL ENGINEERING 75000
## 3 METALLURGICAL ENGINEERING 73000
## 4 NAVAL ARCHITECTURE AND MARINE ENGINEERING 70000
## 5 CHEMICAL ENGINEERING 65000
## 6 NUCLEAR ENGINEERING 65000

```

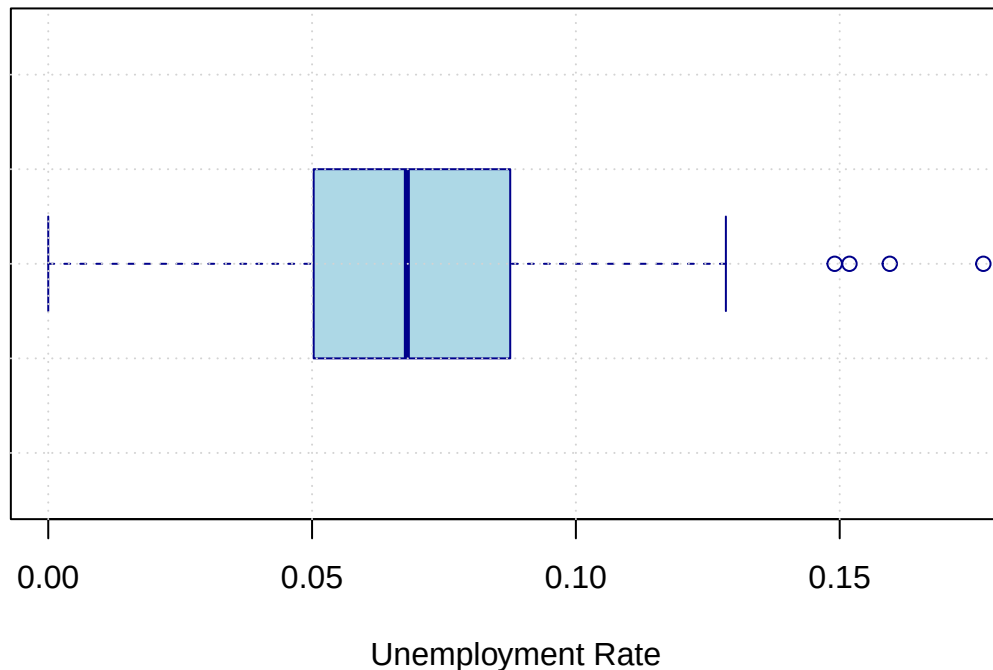
The observed outliers for the Median Salary have majors that are reputed to high-paying jobs. Hence, it can be assumed that the data is accurate. Since, the intention of this analysis is to evaluate the true variation in salaries across all majors, the data of these outliers will be retained in the dataset.

```

boxplot(indata_cleaned$Unemployment_rate,
        main = "Box Plot of Unemployment Rate by Major",
        xlab = "Unemployment Rate",
        col = "lightblue",
        border = "darkblue",
        horizontal = TRUE)
grid()

```

Box Plot of Unemployment Rate by Major



Unemployment Rate

```
unemployment_outliers <- boxplot.stats(indata_cleaned$Unemployment_rate)$out  
unemployment_outlier_majors <- indata_cleaned[indata_cleaned$Unemployment_rate %in% unemployment_outliers,  
unemployment_outlier_majors
```

```
## # A tibble: 4 x 2  
##   Major                Unemployment_rate  
##   <chr>                <dbl>  
## 1 NUCLEAR ENGINEERING 0.177  
## 2 COMPUTER NETWORKING AND TELECOMMUNICATIONS 0.152  
## 3 PUBLIC ADMINISTRATION 0.159  
## 4 CLINICAL PSYCHOLOGY 0.149
```

Nuclear Engineering and Public Administration are fields that have fewer job opportunities and this data reflects the same. Computer Networking and Clinical Psychology are majors that are heavily pursued but with limited opportunities, hence bringing about a higher rate of unemployment. Hence, these outliers are justified and shall be retained in the dataset.

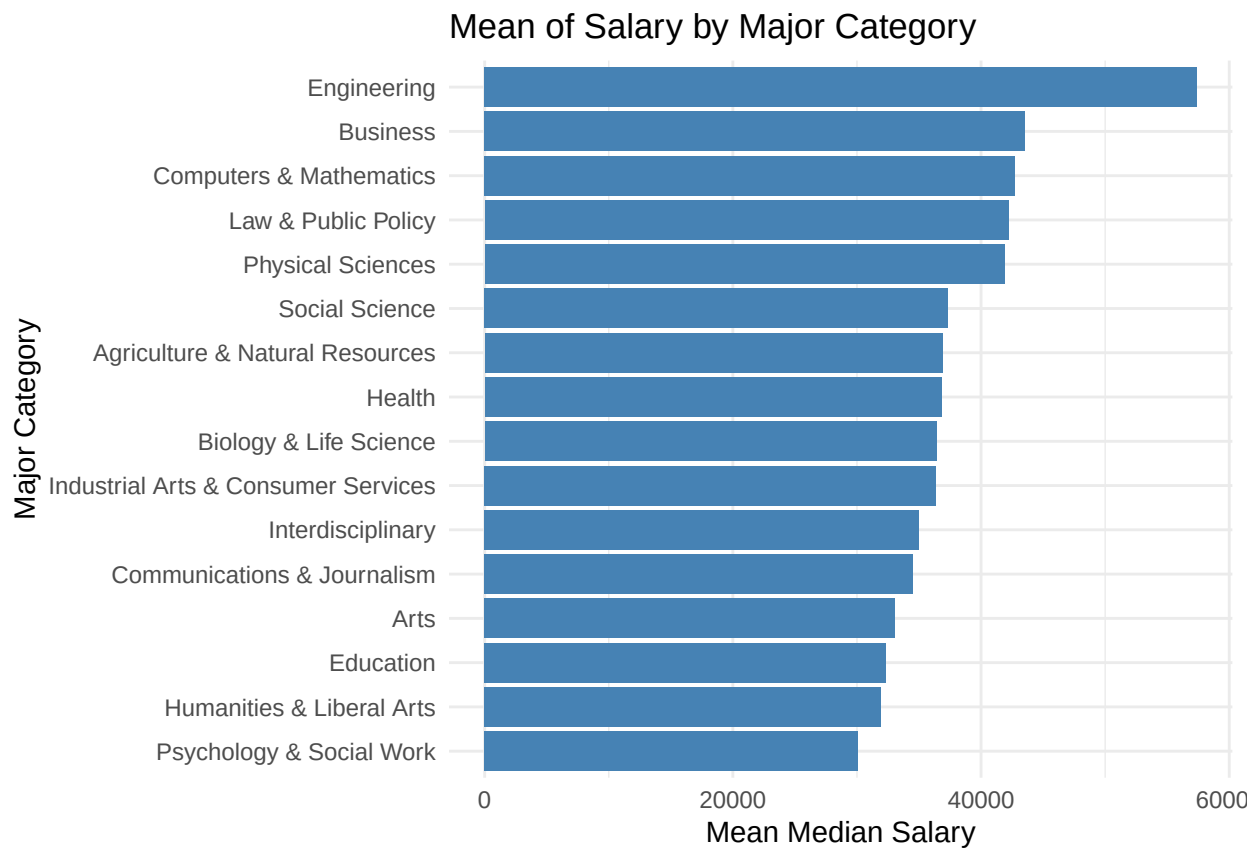
Exploratory Data Analysis

Pay Scale Evaluation of jobs after graduation

In this section, we observe the mean salary of students after graduation in different fields.

```
category_mean_salary <- indata_cleaned %>%
  group_by(Major_category) %>%
  summarize(mean_salary = mean(Median, na.rm = TRUE))

ggplot(category_mean_salary, aes(x = reorder(Major_category, mean_salary), y = mean_salary)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  coord_flip() +
  labs(title = "Mean of Salary by Major Category",
       x = "Major Category",
       y = "Mean Median Salary") +
  theme_minimal()
```



```
box_stats <- boxplot.stats(category_mean_salary$mean_salary)

q1_val <- box_stats$stats[2]
q3_val <- box_stats$stats[4]
print(paste("The Q1 value of mean salary (USD) in each field is:", q1_val))
```

```
## [1] "The Q1 value of mean salary (USD) in each field is: 33781.25"
```

```
print(paste("The Q3 value of mean salary (USD) in each field is:", q3_val))
```

```
## [1] "The Q3 value of mean salary (USD) in each field is: 42045"
```

```
lower_paying_majors <- category_mean_salary[category_mean_salary$mean_salary <= q1_val, c("Major_category", "mean_salary")]
higher_paying_majors <- category_mean_salary[category_mean_salary$mean_salary >= q3_val, c("Major_category", "mean_salary")]
```

Fields with mean salary lower than the first quartile (Q1):

```
print(lower_paying_majors)
```

```
## # A tibble: 4 x 2
##   Major_category      mean_salary
##   <chr>              <dbl>
## 1 Arts                33062.
## 2 Education            32350
## 3 Humanities & Liberal Arts  31913.
## 4 Psychology & Social Work  30100
```

Fields with mean salary greater than the third quartile (Q3):

```
print(higher_paying_majors)
```

```
## # A tibble: 4 x 2
##   Major_category      mean_salary
##   <chr>              <dbl>
## 1 Business            43538.
## 2 Computers & Mathematics  42745.
## 3 Engineering          57383.
## 4 Law & Public Policy    42200
```

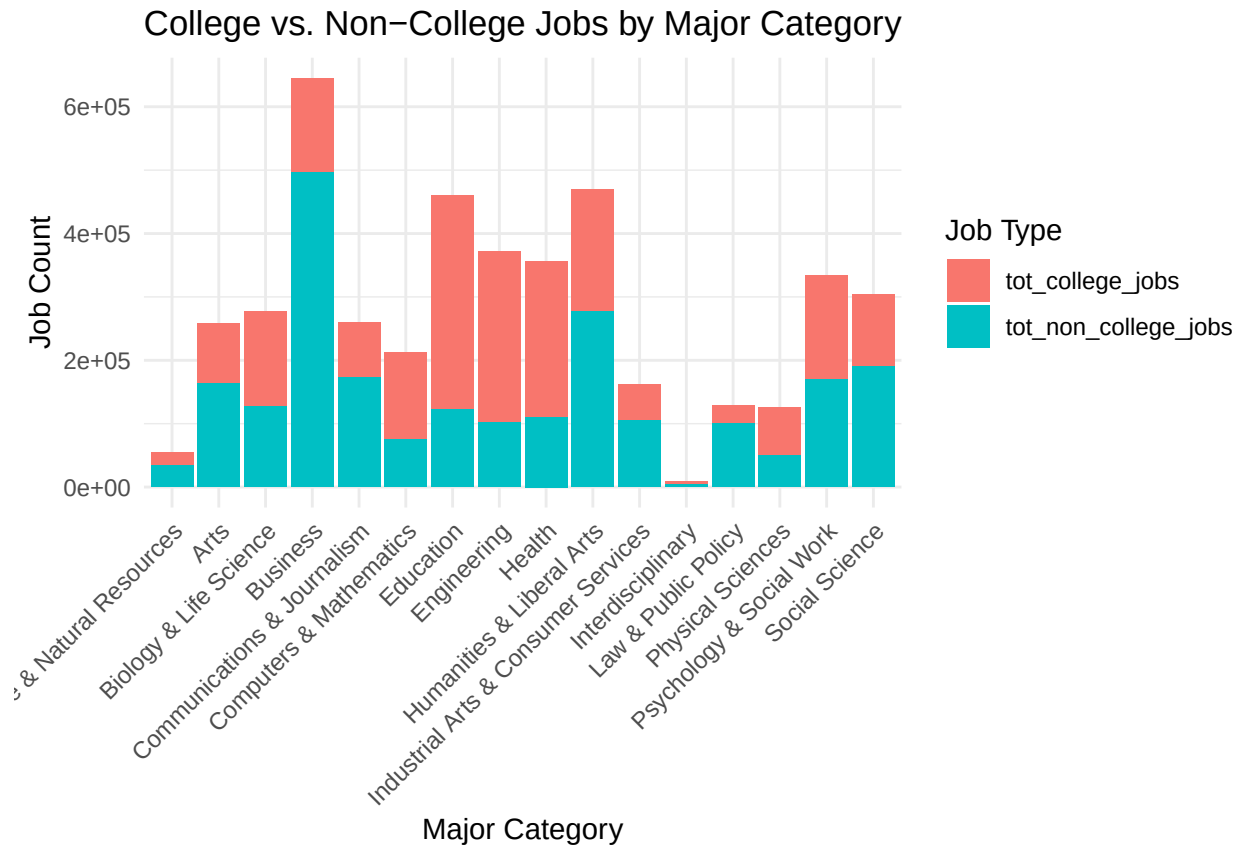
Relevance of Bachelors Degree for jobs after graduation

In this section, we compare the quantity of jobs that require a bachelors degree with the quantity of jobs that do not. This analysis will give us insight into the relevance that Bachelors degree holds in employment.

```
job_distribution <- indata_cleaned %>%
  group_by(Major_category) %>%
  summarize(tot_college_jobs = sum(College_jobs, na.rm = TRUE), tot_non_college_jobs = sum(Non_college_jobs, na.rm = TRUE))

job_distribution_long <- job_distribution %>% pivot_longer(cols = c(tot_college_jobs, tot_non_college_jobs),
  names_to = "Job_Type",
  values_to = "Job_Count")

ggplot(job_distribution_long, aes(x = Major_category, y = Job_Count, fill = Job_Type)) +
  geom_bar(stat = "identity") +
  labs(title = "College vs. Non-College Jobs by Major Category",
    x = "Major Category",
    y = "Job Count",
    fill = "Job Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

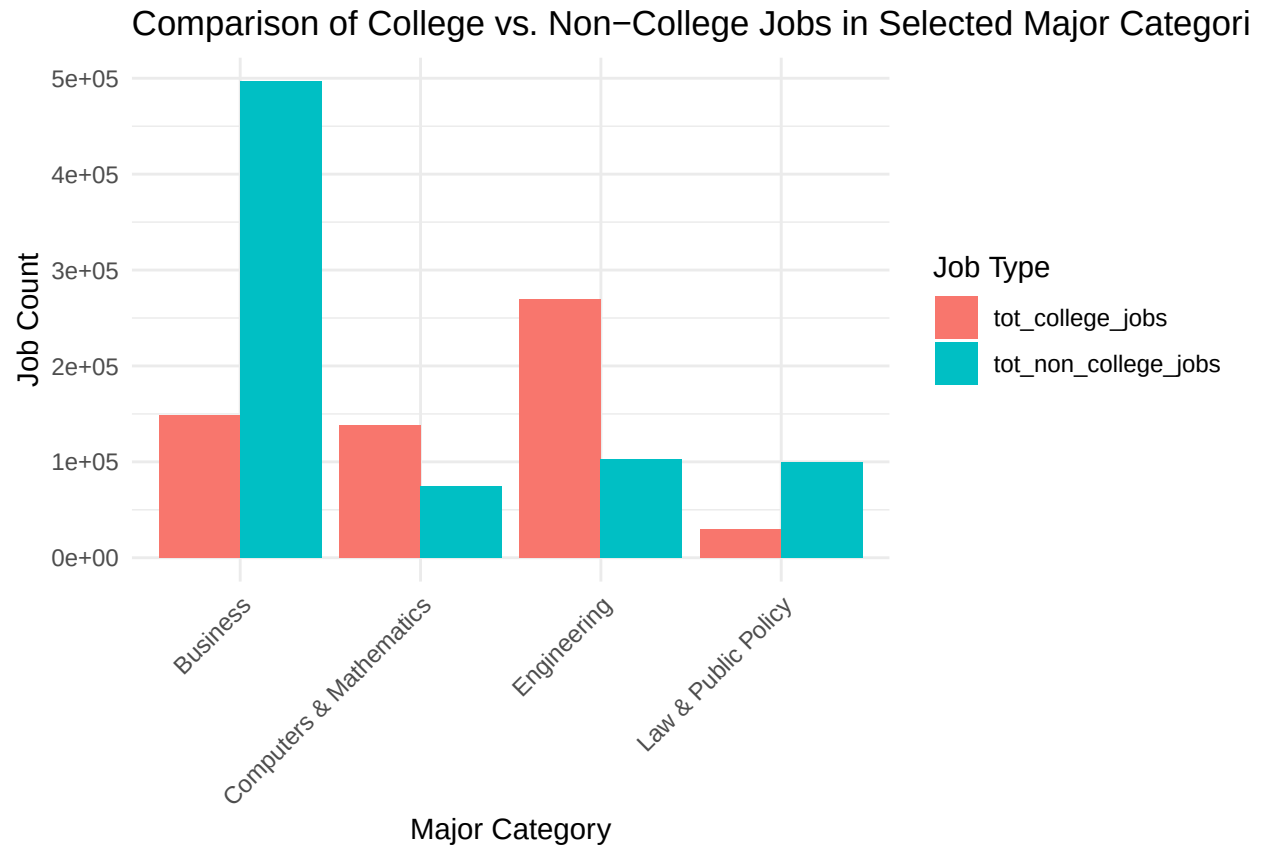


There is a clear indicator that in fields like Arts, Business, Humanities & Liberal Arts, Communications & Journalism, the jobs that do not require a Bachelors degree is distinctly higher than the ones that do.

Below is a similar distribution of jobs for fields that have been previously identified to provide higher paying job opportunities after graduation.

```
job_distribution_high_salary_long <- job_distribution_long[job_distribution_long$Major_category %in% high_salary_categories]

ggplot(job_distribution_high_salary_long, aes(x = Major_category, y = Job_Count, fill = Job_Type)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Comparison of College vs. Non-College Jobs in Selected Major Categories",
       x = "Major Category",
       y = "Job Count",
       fill = "Job Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



In the above plot, we see that in Business, and Law & Public Policy, non-college jobs are higher. However, in Engineering and Computer & Mathematics, we observe that more jobs demand for bachelor's degree.

Pay Scale Range within Fields

Here, we look at the distribution (Q1 and Q3 values of salaries) of Pay Scale within each of the fields.

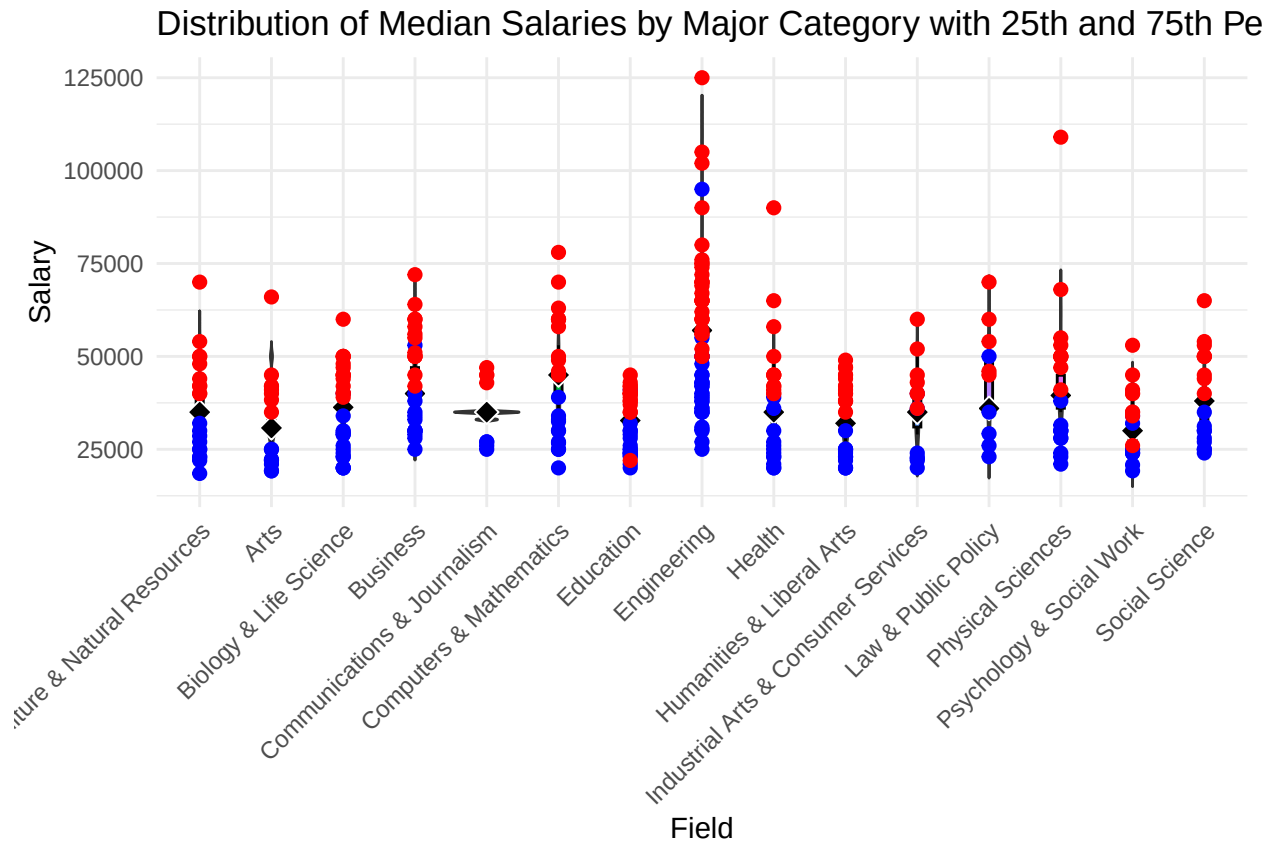
```
job_pay <- indata_cleaned %>%
  group_by(Major_category) %>%
  summarize(lower_pay_mean = mean(P25th, na.rm = TRUE), higher_pay_mean = mean(P75th, na.rm = TRUE))

salary_data <- indata_cleaned %>%
  select(Major_category, Median, P25th, P75th)

salary_data_violin <- salary_data %>%
  group_by(Major_category) %>%
  filter(n() >= 2)

ggplot(salary_data_violin, aes(x = Major_category, y = Median, fill = Major_category)) +
  geom_violin(trim = FALSE, drop = FALSE) +
  geom_boxplot(width = 0.1, outlier.shape = NA, color = "black") +
  stat_summary(fun = median, geom = "point", shape = 23, size = 3, color = "white", fill = "black") +
  geom_point(aes(y = P25th), shape = 21, size = 2, color = "blue", fill = "blue") +
  geom_point(aes(y = P75th), shape = 21, size = 2, color = "red", fill = "red") +
  labs(title = "Distribution of Median Salaries by Major Category with 25th and 75th Percentiles",
```

```
x = "Field",
y = "Salary",
fill = "Major Category") +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "none")
```



As depicted in the above plot, red dots correspond to the third quartile salary for graduates from majors belonging to the field. Blue dots in the plot represent the first quartile salary for graduates from majors of a field. Black dots are the median salary points for each major category, visually emphasizing the central tendency within each distribution.

It is clear from the plot that engineering field offers higher pay scale than rest of the fields.

Employment Rate

In this section, we plot the employment rate in each field.

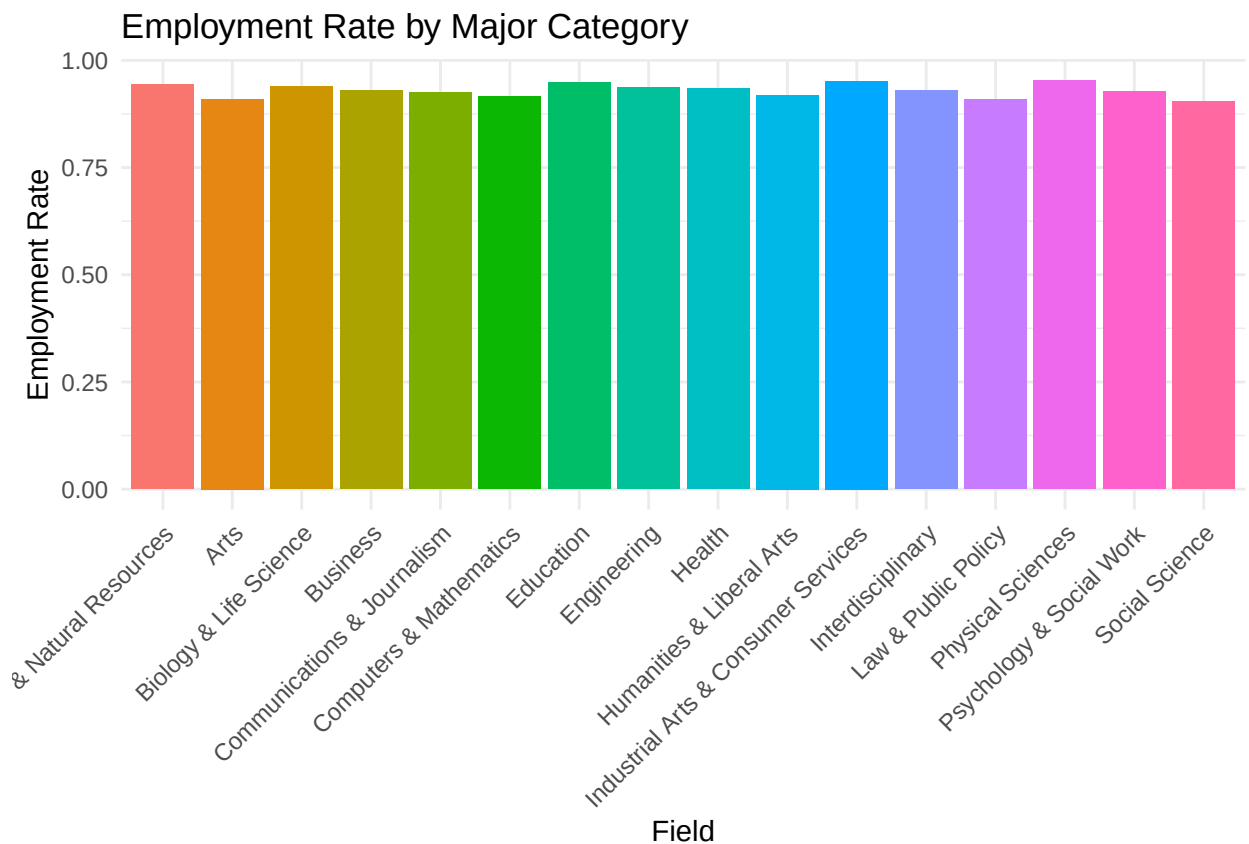
```
employmentData <- indata_cleaned %>%
mutate(employment_rate = 1 - Unemployment_rate)

head(employmentData$employment_rate)
```

```
## [1] 0.9816195 0.8827586 0.9759036 0.9498747 0.9389023 0.8227736
```

```
employmentData <- employmentData %>%
  group_by(Major_category) %>%
  summarize(employment_rate_mean = mean(employment_rate, na.rm = TRUE))

ggplot(employmentData, aes(x = Major_category, y = employment_rate_mean, fill = Major_category)) +
  geom_bar(stat = "identity") +
  labs(title = "Employment Rate by Major Category",
       x = "Field",
       y = "Employment Rate") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```



```
least_employment_categories <- employmentData %>%
  arrange(employment_rate_mean) %>%
  slice_head(n = 5)

print(least_employment_categories)
```

```
## # A tibble: 5 x 2
##   Major_category      employment_rate_mean
##   <chr>              <dbl>
## 1 Social Science      0.904
## 2 Law & Public Policy  0.909
## 3 Arts                0.910
```


## 4 Computers & Mathematics	0.916
## 5 Humanities & Liberal Arts	0.919

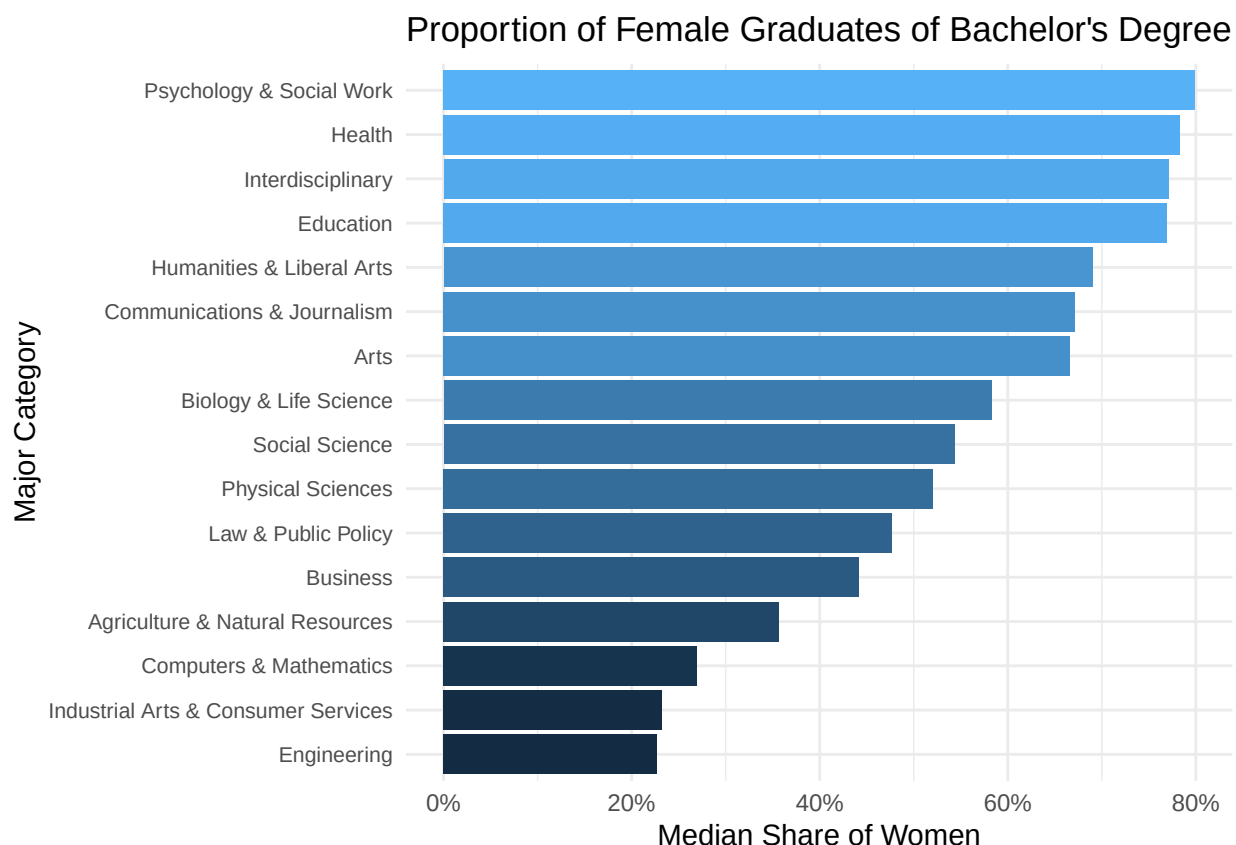
While most of the majors seem to have good overall employment rate, the fields with the least employment rate are shown above.

Female Population in Majors

Below plot shows the proportion of females in the major fields. It is observed that in fields like Psychology & Social Work, Health etc have >75% population to be females while fields like Engineering, Industrial Arts & Consumer Services have significantly lesser female population.

```
category_summary <- indata_cleaned %>%
  group_by(Major_category) %>%
  summarize(median_share_women = median(ShareWomen, na.rm = TRUE))

# Create a bar chart to visualize the median share of women in each major category
ggplot(category_summary, aes(x = reorder(Major_category, median_share_women), y = median_share_women, fill = "red")) +
  geom_bar(stat = "identity") +
  labs(title = "Proportion of Female Graduates of Bachelor's Degree",
       x = "Major Category",
       y = "Median Share of Women") +
  scale_y_continuous(labels = scales::percent) + # Format y-axis as percentage
  coord_flip() + # Flip for horizontal bars
  theme_minimal() +
  theme(legend.position = "none", # Remove legend
        axis.text.y = element_text(size = 8))
```



Inferences

1. Employment Outcomes by Degree Major Type - Hypothesis Testing

Null Hypothesis (H_0) : The mean employment rate is independent on the major category.

Alternative Hypothesis (H_1): The mean employment rate is dependent on the major category.

Test Type: One-way ANOVA (Analysis of Variance) can be used to assess the dependency of employment rate across different major categories.

Test Statistic: In this hypothesis testing, F-statistic is used as the test statistic. F-statistic is appropriate here because it captures the variance of the mean employment rate amongst the major categories and within the major categories too.

Reference Distribution: The reference distribution for F-statistic in ANOVA testing is F-distribution characterized by two degrees and denoted as $F(df_1, df_2)$. The degrees of freedom are calculated as below:

1. df_1 : This value captures the degrees of freedom between the major categories.

$$df_1 = k-1,$$

where k = number of unique major categories.

In this scenario, the value of $df_1 = 16-1 = 15$

2. df_2 : This value captures the degrees of freedom within the major categories.
 $df_2 = N - k$,
 where k = number of unique major categories
 and N = Total number of observations.
 In this scenario, the value of $df_2 = 173 - 16 = 157$

Hence, the reference distribution for the F-statistic is F-distribution with $F(15, 157)$.

Rejection Criteria: The criteria to reject the null hypothesis is if p-value of ANOVA test is less than the significance level. Here, we set the significance value = 0.05, meaning we are willing to accept a 5% chance of concluding that the mean employment rate differs by major category when it actually does not.

```
test1_data <- indata_cleaned %>% mutate(Employment_Rate = 1-Unemployment_rate)
anova_result <- aov(Employment_Rate ~ Major_category, data = test1_data)

# Display ANOVA summary
summary(anova_result)
```

ANOVA Test:

```
##              Df  Sum Sq   Mean Sq F value    Pr(>F)
## Major_category  15  0.03379  0.0022529    2.842 0.000591 ***
## Residuals      157  0.12444  0.0007926
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion: The F value is 2.842. A higher F-statistic indicates that the between-group variance is large relative to the within-group variance, which suggests that the differences between groups are not due to random chance. Since the p-value (0.000591) is well below 0.05, we reject the null hypothesis.

There is statistically significant evidence that the mean employment rate differs by major category. In other words, **at least one major category has a mean employment rate that is significantly different from others.** **

2. Identify factors influencing Median Salary - PCA & Linear Regression

In this section, Principal Component Analysis and Linear Regression are performed to reduce the dimensionality of the data and to identify factors that influence the Median salary of a college graduate.

Data Pre-processing Data pre-processing must be completed before moving on with PCA. First, columns unrelated to the analysis, such as Major and Sample_size, are eliminated. A few columns contain information that is already recorded in another column. For example, the values in P25th and P75th are not needed since the dataset already contains the median income, and the columns Men and Women are not needed because the dataset already contains the Total and WomenShare columns. We minimise the dataset to contain just non-redundant information by eliminating these unnecessary columns.

Ensuring the data is entirely numerical would be the next stage. One of the column - Major_Category is crucial to the analysis and is not numeric. Hence, it must be encoded.

```

selected_data <- subset(indata_cleaned, select = -c(Major, Men, Women, Sample_size, P25th, P75th, Employ
one_hot_encoded <- model.matrix(~ Major_category - 1, data = selected_data)
numeric_data <- selected_data%>%select_if(is.numeric)

pca_data <- cbind(numeric_data, one_hot_encoded)

```

```
cor_matrix <- cor(pca_data)
```

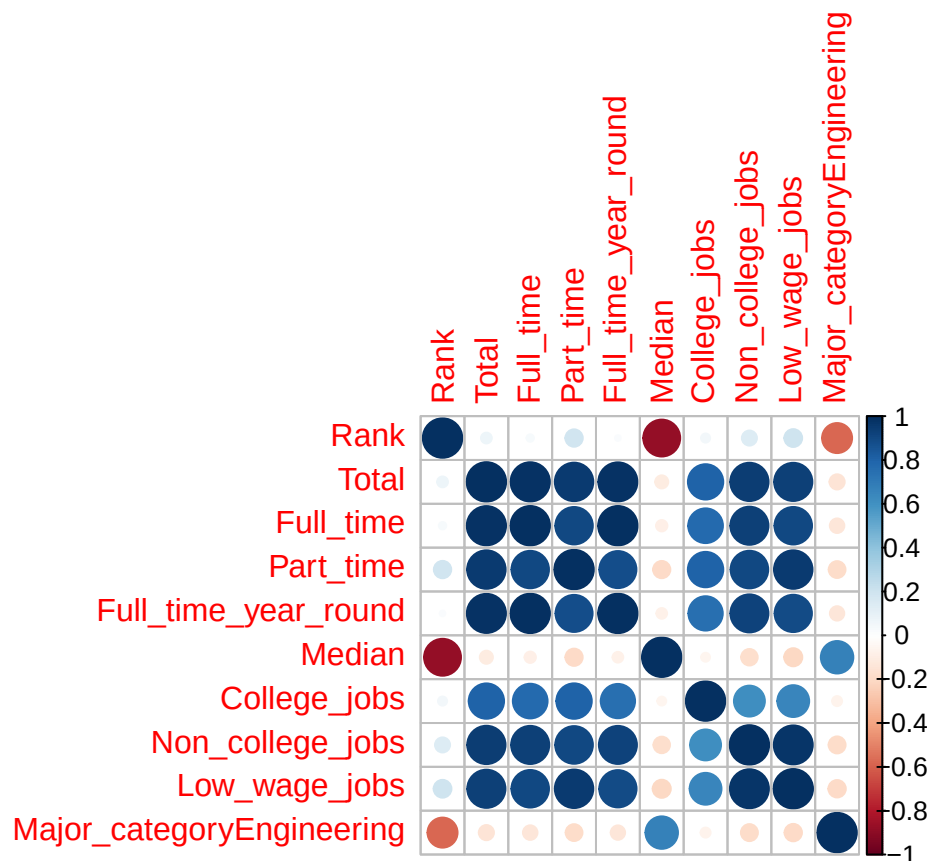
Perform Correlation The correlation matrix is generated with the processed data.

Since, it is difficult to go through a matrix of this length, below is a plot that captures the columns that have variance greater than 0.1.

```

high_variance_data <- pca_data[, apply(pca_data, 2, var) > 0.1]
high_cor_matrix <- cor(high_variance_data)
corrplot(high_cor_matrix, method = "circle")

```



This plot gives us a better understanding of the influence the variables have on one another. Clearly from the plot, we observe multicollinearity.

Principal Component Analysis The Principal Component Analysis is performed after scaling the processed data. The summary and the Scree Plot are generated to understand the principal components better.

```
# Step 3: Standardize the data
scaled_data <- scale(pca_data)
```

```
# Step 4: Perform PCA
pca <- prcomp(scaled_data, center = TRUE, scale. = TRUE)
```

```
# Summary of PCA
summary(pca)
```

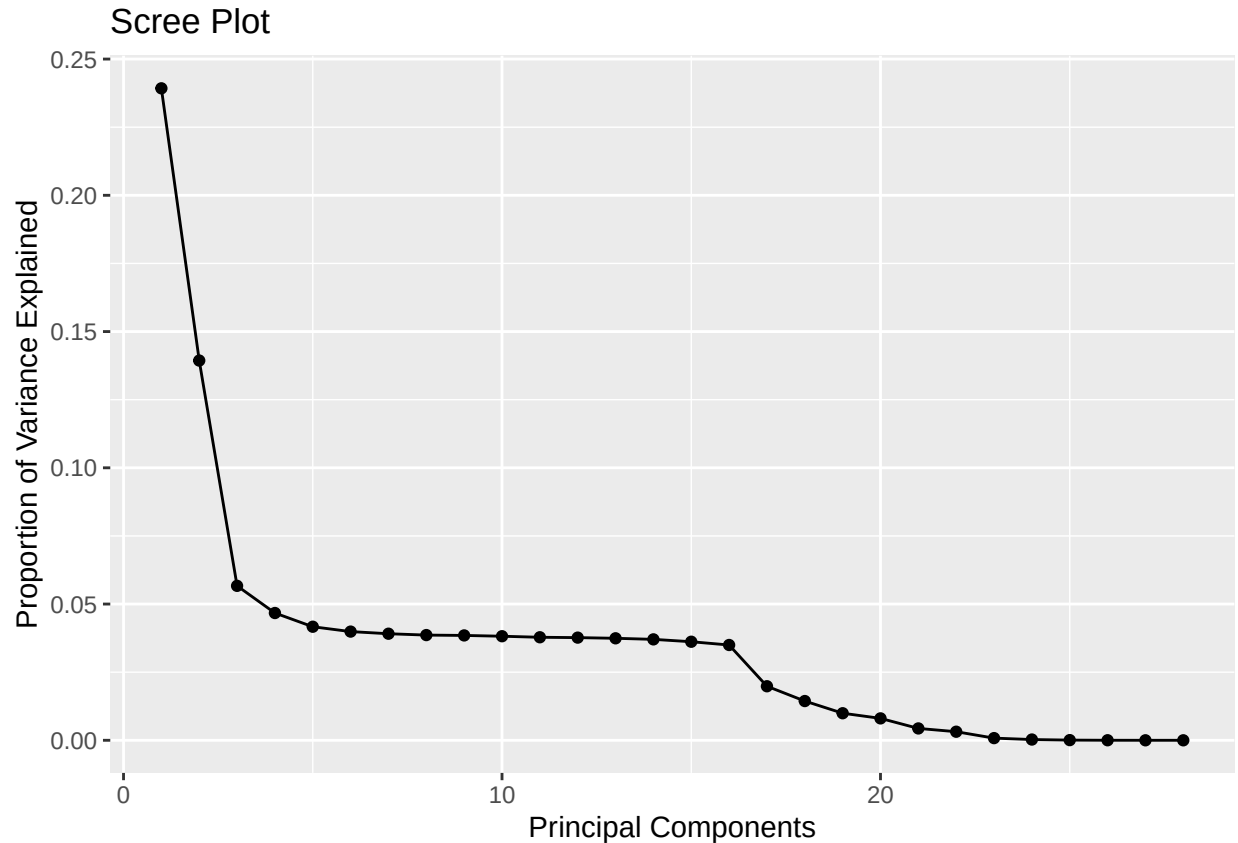
```
## Importance of components:
```

```
##           PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation    2.5883 1.9754 1.25976 1.14364 1.08025 1.05689 1.04649
## Proportion of Variance 0.2393 0.1394 0.05668 0.04671 0.04168 0.03989 0.03911
## Cumulative Proportion 0.2393 0.3786 0.43530 0.48201 0.52369 0.56358 0.60269
##           PC8      PC9      PC10     PC11     PC12     PC13     PC14
## Standard deviation    1.03981 1.03811 1.0341 1.02893 1.02717 1.02380 1.01840
## Proportion of Variance 0.03861 0.03849 0.0382 0.03781 0.03768 0.03743 0.03704
## Cumulative Proportion 0.64131 0.67980 0.7180 0.75580 0.79348 0.83092 0.86796
##           PC15     PC16     PC17     PC18     PC19     PC20     PC21
## Standard deviation    1.00640 0.98954 0.74529 0.6350 0.52747 0.47459 0.34931
## Proportion of Variance 0.03617 0.03497 0.01984 0.0144 0.00994 0.00804 0.00436
## Cumulative Proportion 0.90413 0.93910 0.95894 0.9733 0.98328 0.99132 0.99568
##           PC22     PC23     PC24     PC25     PC26     PC27
## Standard deviation    0.29705 0.1495 0.08908 0.04422 0.02094 1.149e-15
## Proportion of Variance 0.00315 0.0008 0.00028 0.00007 0.00002 0.000e+00
## Cumulative Proportion 0.99883 0.9996 0.99991 0.99998 1.00000 1.000e+00
##           PC28
## Standard deviation    3.875e-16
## Proportion of Variance 0.000e+00
## Cumulative Proportion 1.000e+00
```

```
scree_plot <- qplot(1:length(pca$sdev), pca$sdev^2 / sum(pca$sdev^2),
                    geom = 'line') +
  geom_point() +
  labs(title = "Scree Plot", x = "Principal Components", y = "Proportion of Variance Explained")
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
print(scree_plot)
```



From the summary of the PCA and Scree Plot, it is evident that the 90% of the variability is captured by the first 15 components. It is also observed that the components beyond PC22 add almost 0 variability to the dataset.

Based on the principal components generated, it is important to identify the top factors that influence the principal components. Below are the top factors influencing the first 15 components.

```
loading_matrix <- pca$rotation

important_predictors <- apply(loading_matrix, 2, function(x) names(x[order(-abs(x))][1:3]))
print(important_predictors[, 1:15])
```

```
##      PC1          PC2          PC3
## [1,] "Total"      "mean_womenshare" "Unemployment_rate"
## [2,] "Part_time"  "Rank"      "Major_categoryEducation"
## [3,] "Low_wage_jobs" "Median"      "Major_categorySocial Science"
##      PC4
## [1,] "Major_categoryHealth"
## [2,] "Major_categoryIndustrial Arts & Consumer Services"
## [3,] "Major_categoryAgriculture & Natural Resources"
##      PC5
## [1,] "Major_categoryBusiness"
## [2,] "Major_categoryEngineering"
## [3,] "Major_categoryHumanities & Liberal Arts"
##      PC6
## [1,] "Major_categoryComputers & Mathematics"
## [2,] "Major_categoryEducation"
```

```

## [3,] "Major_categoryHumanities & Liberal Arts"
##      PC7
## [1,] "Major_categoryBiology & Life Science"
## [2,] "Major_categoryEducation"
## [3,] "Major_categoryBusiness"
##      PC8
## [1,] "Major_categoryHumanities & Liberal Arts"
## [2,] "Major_categoryBiology & Life Science"
## [3,] "Major_categoryPhysical Sciences"
##      PC9
## [1,] "Major_categoryIndustrial Arts & Consumer Services"
## [2,] "Major_categoryHumanities & Liberal Arts"
## [3,] "Major_categoryHealth"
##      PC10
## [1,] "Major_categoryPhysical Sciences"
## [2,] "Major_categoryPsychology & Social Work"
## [3,] "Major_categoryHealth"
##      PC11
## [1,] "Major_categoryPsychology & Social Work"
## [2,] "Major_categoryIndustrial Arts & Consumer Services"
## [3,] "Major_categoryAgriculture & Natural Resources"
##      PC12
## [1,] "Major_categorySocial Science"
## [2,] "Major_categoryArts"
## [3,] "Major_categoryAgriculture & Natural Resources"
##      PC13
## [1,] "Major_categoryIndustrial Arts & Consumer Services"
## [2,] "Major_categoryAgriculture & Natural Resources"
## [3,] "Major_categoryArts"
##      PC14
## [1,] "Major_categoryLaw & Public Policy"
## [2,] "Major_categoryCommunications & Journalism"
## [3,] "Major_categoryBusiness"
##      PC15
## [1,] "Major_categoryInterdisciplinary"
## [2,] "Major_categoryCommunications & Journalism"
## [3,] "Major_categoryLaw & Public Policy"

```

Identify Principal Components influencing Median Salary Linear Regression is performed to identify the principal components that significantly influence Median Salary. After experimenting with several principal component combinations, it was found that PC1, PC2, PC3, PC4, and PC6 appear to have the greatest influence on Median Salary and in obtaining the optimal combination of R-squared value and F-statistic.

```

# Add PCA scores to the dataset
pca_scores <- as.data.frame(pca$x)
pca_scores <- cbind(pca_data$Median, pca_scores)
colnames(pca_scores)[1] <- "MedianSalary"

# Fit regression model using the first three principal components
pca_regression <- lm(MedianSalary ~ PC1 + PC2 + PC3 + PC4 + PC6 , data = pca_scores)

# Display regression summary

```

```
summary(pca_regression)
```

```
##
## Call:
## lm(formula = MedianSalary ~ PC1 + PC2 + PC3 + PC4 + PC6, data = pca_scores)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13119  -2193      90    1623   35189
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  40151.4      349.7  114.807 < 2e-16 ***
## PC1           1415.8       135.5   10.448 < 2e-16 ***
## PC2          -4644.9       177.6  -26.160 < 2e-16 ***
## PC3           1926.5       278.4    6.919 9.25e-11 ***
## PC4          -2253.7       306.7   -7.348 8.49e-12 ***
## PC6            882.5       331.9    2.659  0.0086 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4600 on 167 degrees of freedom
## Multiple R-squared:  0.8438, Adjusted R-squared:  0.8392
## F-statistic: 180.5 on 5 and 167 DF,  p-value: < 2.2e-16
```

The model has a p-value $< 2e-16$ indicating that the model is statistically significant. Based on the summary of the model, it can be observed that PC2 and PC4 negatively impact the median salary while PC1, PC3 and PC6 positively impact the median salary.

Conclusion Conclusions for the above analysis can be drawn by studying both the linear regression model summary and the important predictors of principal components.

The top influencing factors of PC1, PC3, and PC6 are **Total graduates, Part_time Jobs, Low wage jobs, Unemployment_rate, Major_categories - Education, Social Science, Computer & Mathematics, Humanities & Liberal Arts** which in turn positively impact the Median Salary.

The top influencing factors for PC2 and PC4 are **WomenShare, Rank, Major_categories - Health, Industrial Arts & Consumer Services, Agriculture & Natural Resources** which negatively impact the Median salary of the Graduate.

Conclusion

The analysis presented thoroughly examines the relationship between earning a bachelor's degree and financial outcomes, using data from FiveThirtyEight based on the American Community Survey (2010-2012). Through this analysis, we have successfully explored various facets of this relationship, including employment rates, salary variations across major categories, and the influence of factors like gender distribution and the relevance of a degree in different job markets.

The core findings of the analysis can be generalized into three key points:

1. **Earning Potential Varies Significantly by Major:** The choice of major has a substantial impact on potential earnings. Fields like Engineering, Computers & Mathematics, Business, and Law & Public Policy consistently demonstrate higher median salaries than others, such as Arts, Education, Humanities & Liberal Arts, and Psychology & Social Work. This suggests that certain fields offer a more favorable return

on investment in terms of financial outcomes.

2. Bachelor's Degrees are not Universally Essential, but hold Significant Value in Specific Fields: While some industries have a large proportion of jobs that do not necessitate a bachelor's degree, fields like Engineering and Computers & Mathematics exhibit a strong demand for degree holders. This reinforces the notion that a bachelor's degree can be a crucial requirement for accessing higher-paying positions in certain technical and specialized fields.

3. Multiple Factors Influence Salary Outcomes: By employing statistical techniques like principal component analysis (PCA), and linear regression, factors that contribute to variations in salary are identified. These factors include and are not limited to the total number of graduates in a major, the prevalence of part-time and low-wage jobs within a field, the unemployment rate for specific majors, and the proportion of women in a major.

Hence, while a bachelor's degree is not a guaranteed path to financial success, it can significantly enhance earning potential, particularly in fields with high demand for skilled graduates. The decision to pursue a bachelor's degree should involve careful consideration of individual career aspirations, the job market dynamics within specific industries, and an understanding of the various factors that can impact salary outcomes.

While the analysis provides great insights into the topic, it is important to note the limitations:

1. The dataset used is based on a specific time period (2010-2012), and economic conditions and job market trends can fluctuate over time.
2. Further research could explore longitudinal data to understand salary trends and the evolving value of different majors.
3. Additionally, the analysis focuses primarily on quantitative data. Conducting qualitative studies, such as interviews with graduates from various fields, could provide valuable insights into the lived experiences and career paths of individuals with different degrees.